

Simulation Thinking

Jayanth Raghothama

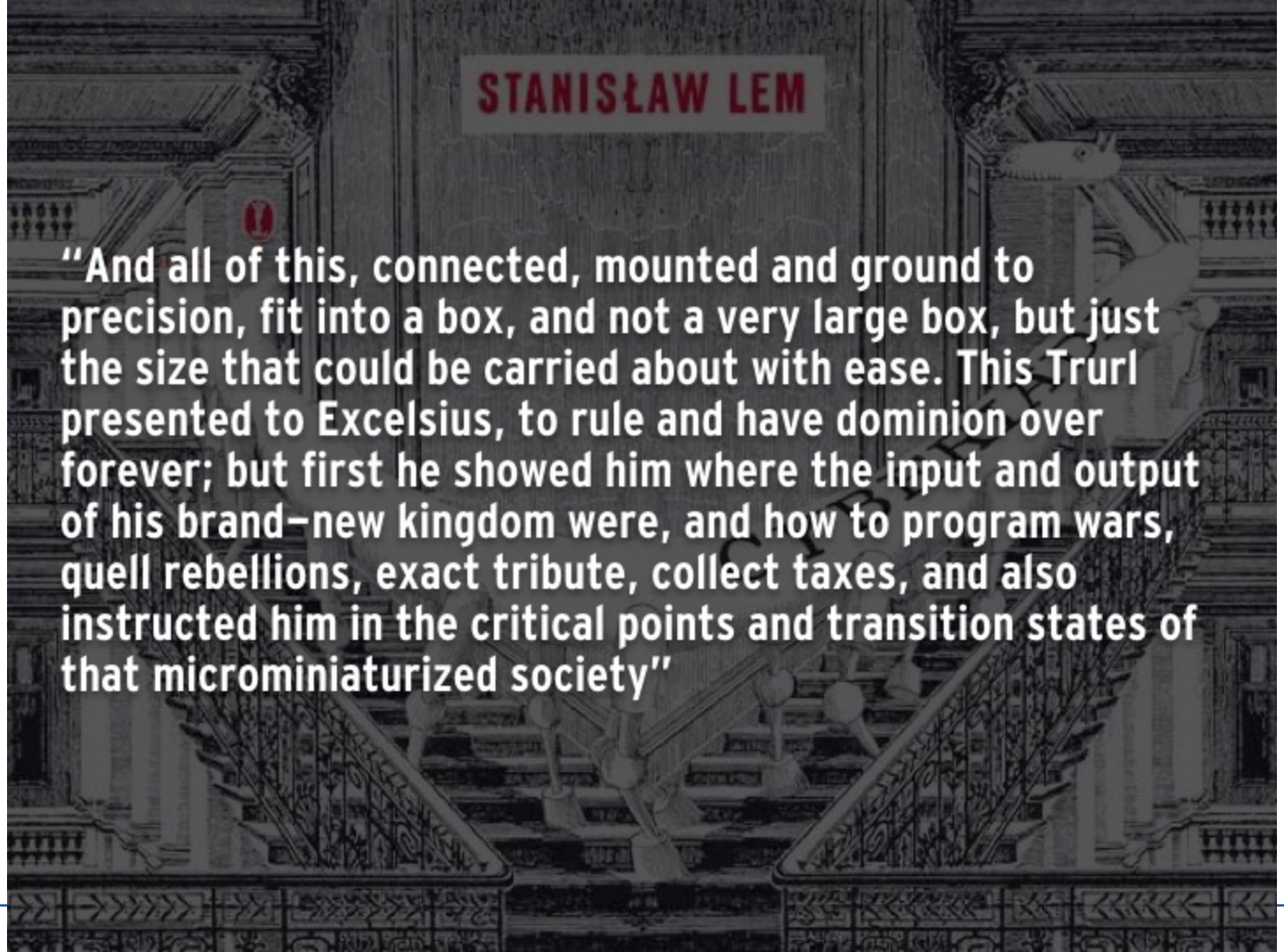
Assistant Professor

jayanthr@kth.se



The Seventh Sally or How Trurl's Own Perfection Led to No Good

From Cyberiad by Stanislaw Lem (1965)

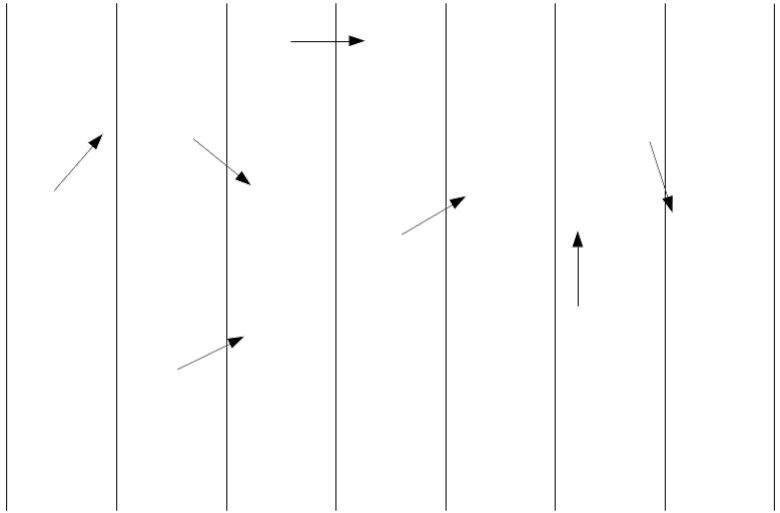

STANISŁAW LEM
"And all of this, connected, mounted and ground to precision, fit into a box, and not a very large box, but just the size that could be carried about with ease. This Trurl presented to Excelsius, to rule and have dominion over forever; but first he showed him where the input and output of his brand-new kingdom were, and how to program wars, quell rebellions, exact tribute, collect taxes, and also instructed him in the critical points and transition states of that microminiaturized society"

What is a simulation?

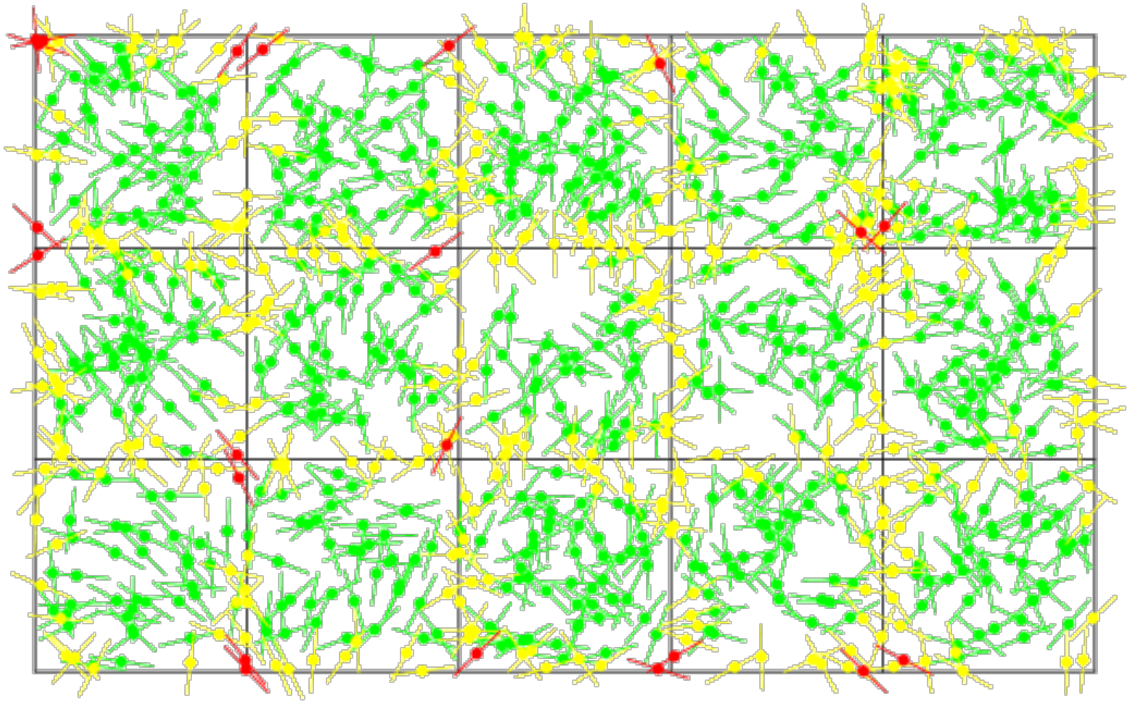
“A simulation imitates one process by another process. In this definition the term “process” refers solely to some object or system whose state changes in time. If the simulation is run on a computer, it is called a computer simulation”

- - Stephan Hartmann (1996)

Buffon Needle Problem



Monte-Carlo Method



Guiness, WW2

- Gosset created the Student's-t distribution, a cornerstone of statistics for decades. Working for Guiness brewery, he combined the Monte Carlo method with analytical techniques to estimate the quality of *stout*
- Stanislaw Ulan, John von Neumann and others use the Monte Carlo method on electronic computers in order to solve certain problems in neutron diffusion that arose in the design of the hydrogen bomb and that were (and still are) analytically intractable



This Lecture

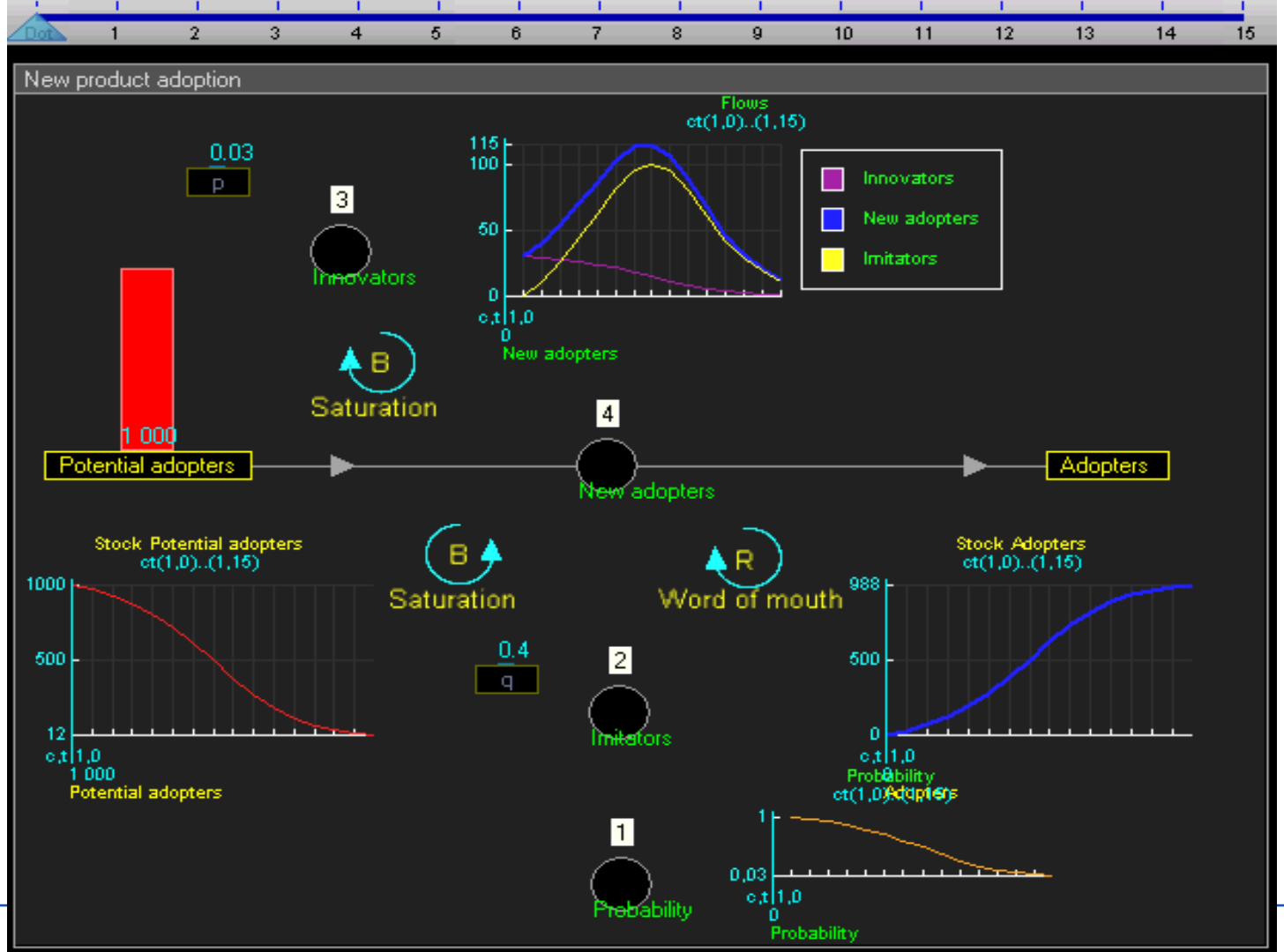
- Simulation Types – An Overview
- Representation
- Phases of a Simulation Study
- Conceptual Modeling

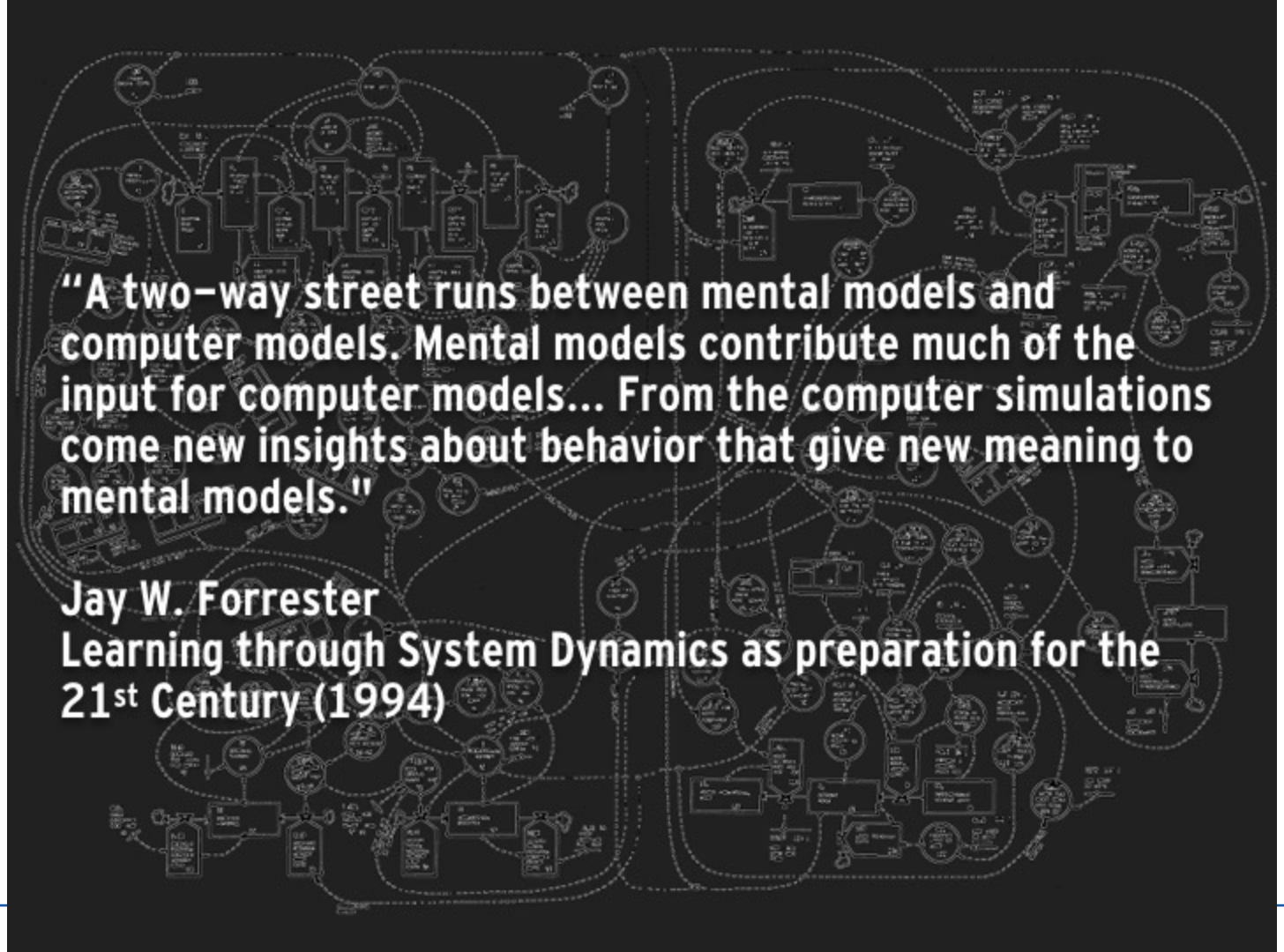
System Type?

- Natural
- Artificial
- Complex Adaptive
- Closed vs Open

Complex Systems

- Emergent
- Chaotic
- Non-Linear
- Sensitive to initial conditions
- Catastrophic Failures
- Human Actors Present


Dot
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New product adoption
Flows
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New adopters
Innovators
New adopters
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Saturation
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Potential adopters
4
New adopters
Adopters
Stock Potential adopters
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Potential adopters
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Probability

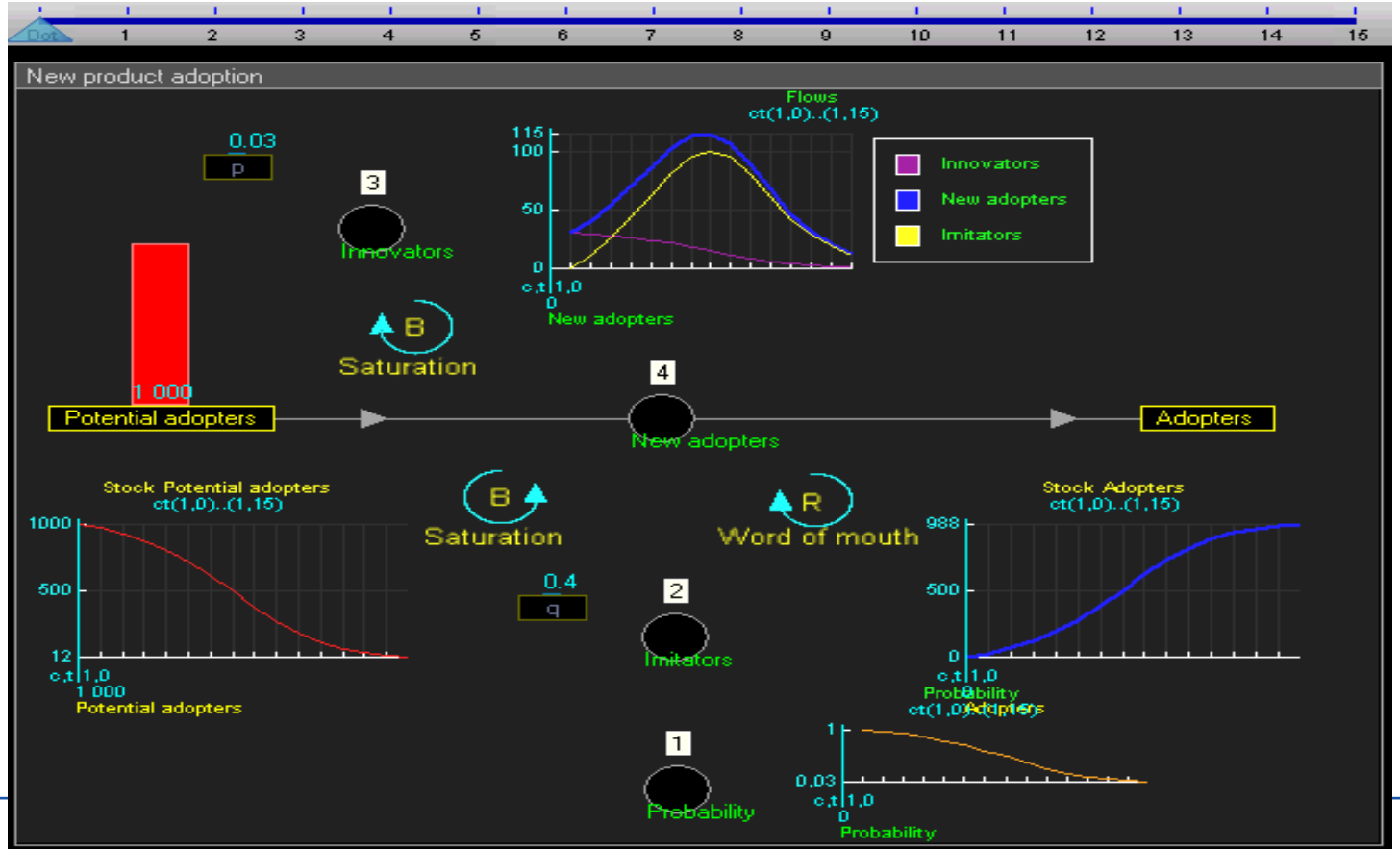

"A two-way street runs between mental models and computer models. Mental models contribute much of the input for computer models... From the computer simulations come new insights about behavior that give new meaning to mental models."
Jay W. Forrester
Learning through System Dynamics as preparation for the 21st Century (1994)

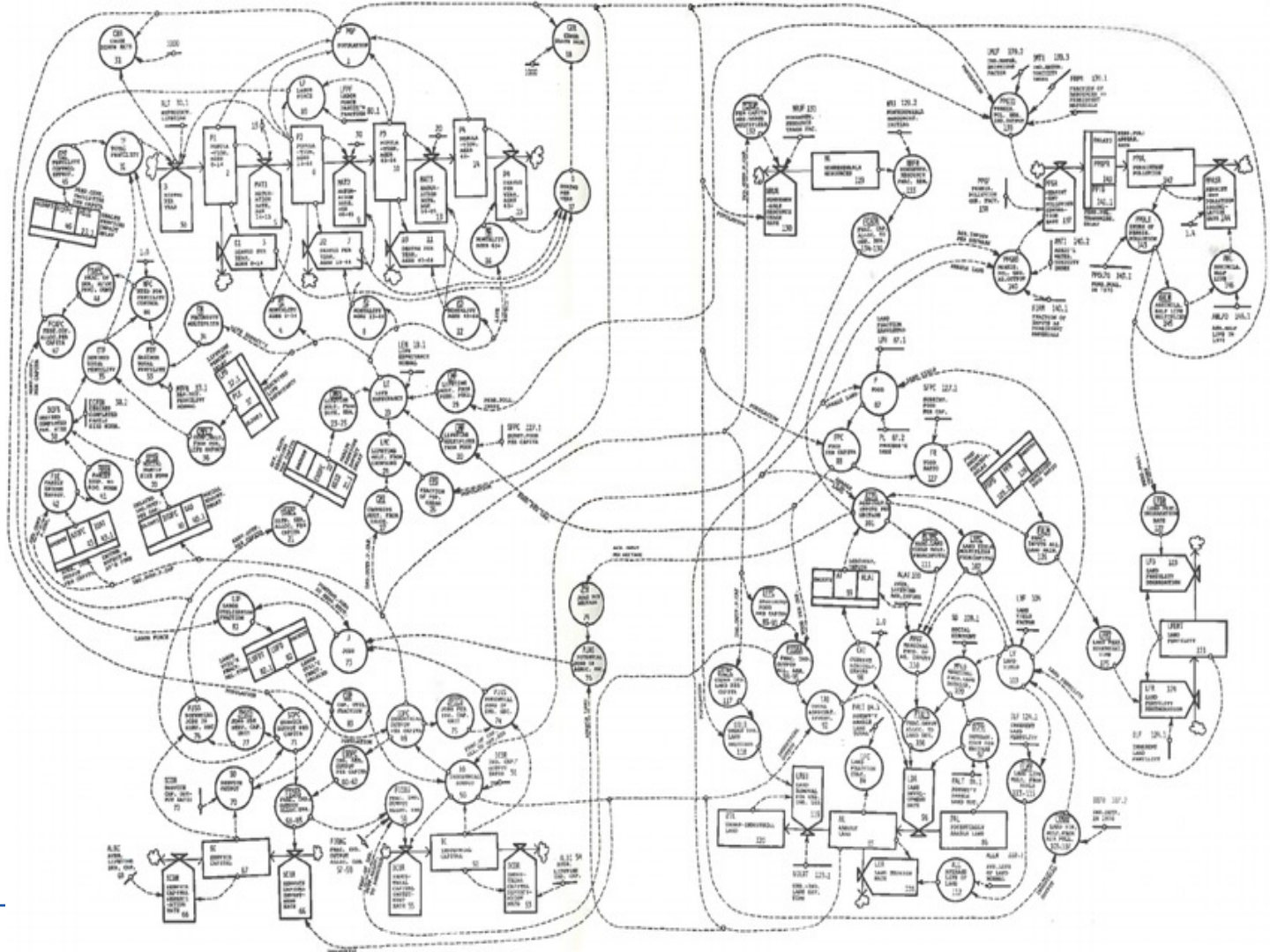


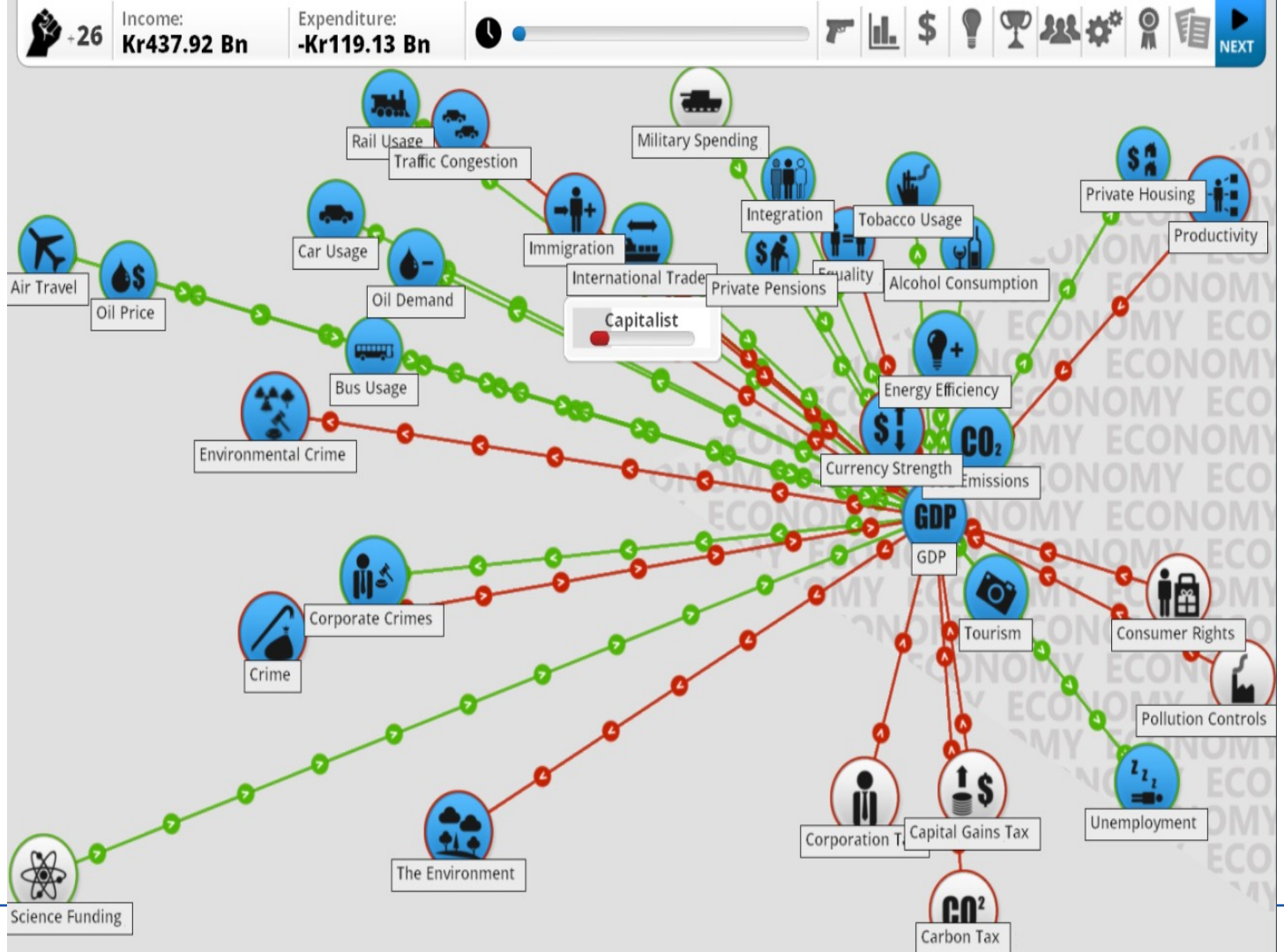
Types

- Based on paradigm, or the type of system
  - System Dynamics
  - ABM
  - DES etc.
  - CFD
  - FEM
  - Hybrid
- Based on computation paradigm
  - Distributed
  - Participatory

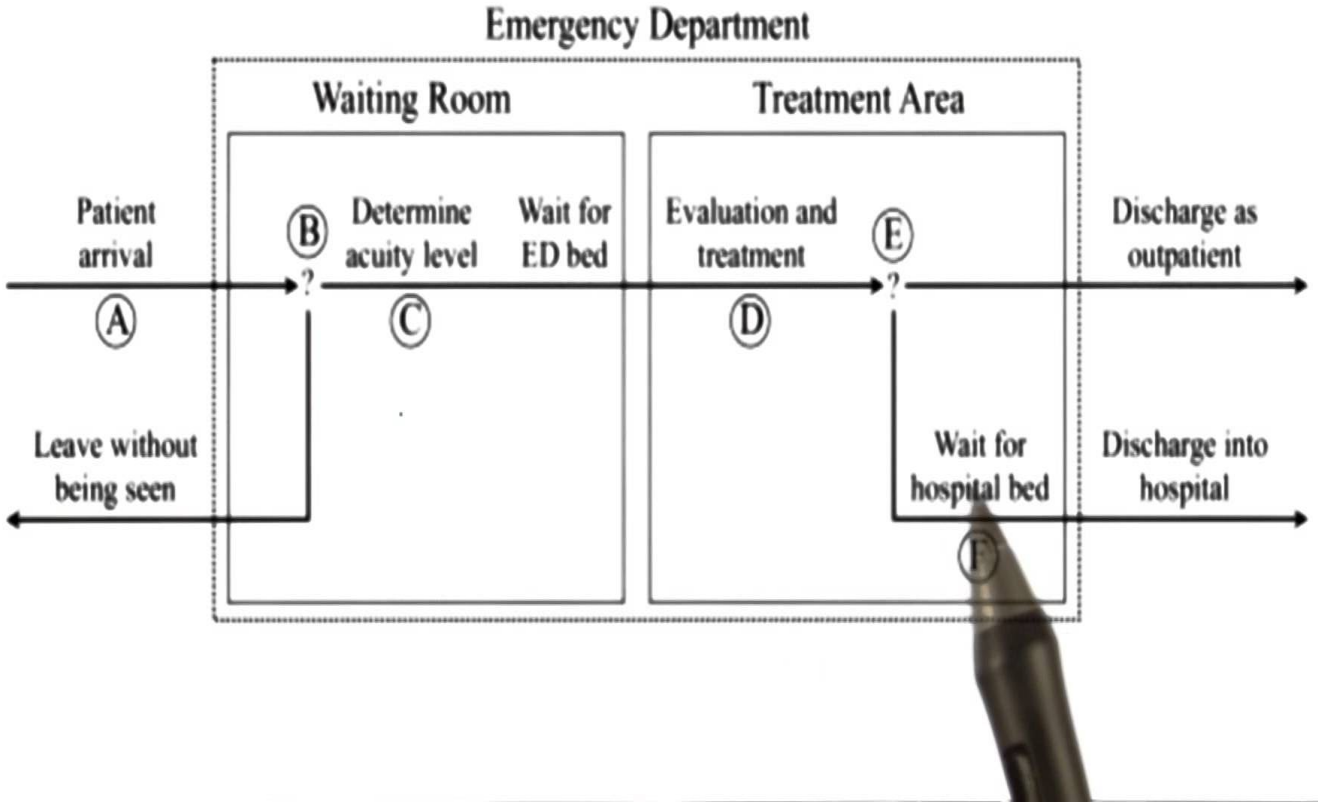
Systems Dynamics





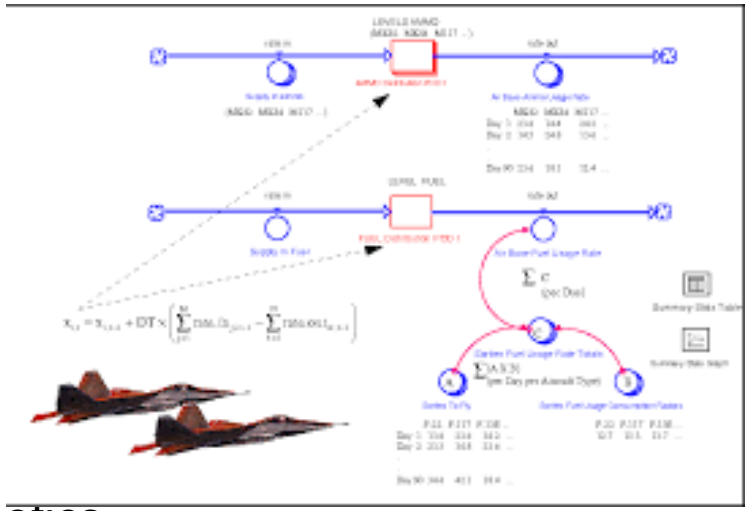

+26
Income:
Kr437.92 Bn
Expenditure:
-Kr119.13 Bn
NEXT
Rail Usage
Traffic Congestion
Military Spending
Private Housing
Productivity
Integration
Tobacco Usage
Car Usage
Immigration
International Trade
Private Pensions
Alcohol Consumption
Air Travel
Oil Price
Oil Demand
Capitalist
Bus Usage
Energy Efficiency
Environmental Crime
Currency Strength
Emissions
GDP
GDP
Corporate Crimes
Tourism
Consumer Rights
Crime
Pollution Controls
Unemployment
Capital Gains Tax
Corporation T
The Environment
Science Funding
Carbon Tax

DISCRETE EVENT SIMULATION



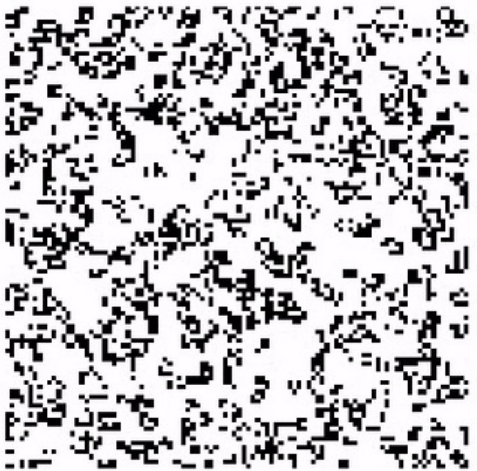
Continuous Simulation

- Model of a physical system that continuously tracks system response according to a set of differential equations
- Flight tracking, water flow, robotics



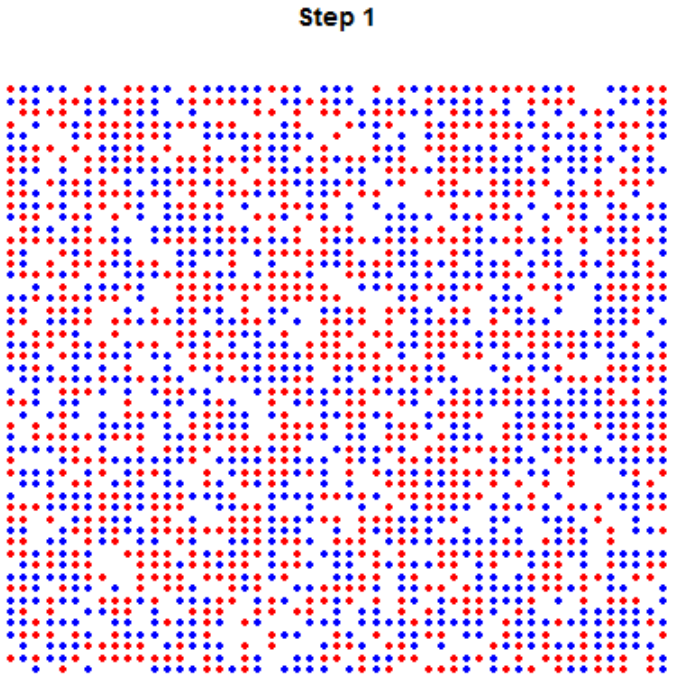
Agent Based Models

- Understand behavioral models
- Simulate decision making
- Entertainment, Games ..



Schelling's Segregation Model

Model of racial segregation in Chicago in 1970's.
He wanted to show that you can get segregation with a low level of preference, for example 30%
Rule: Every step, each cell looks at neighbors, if

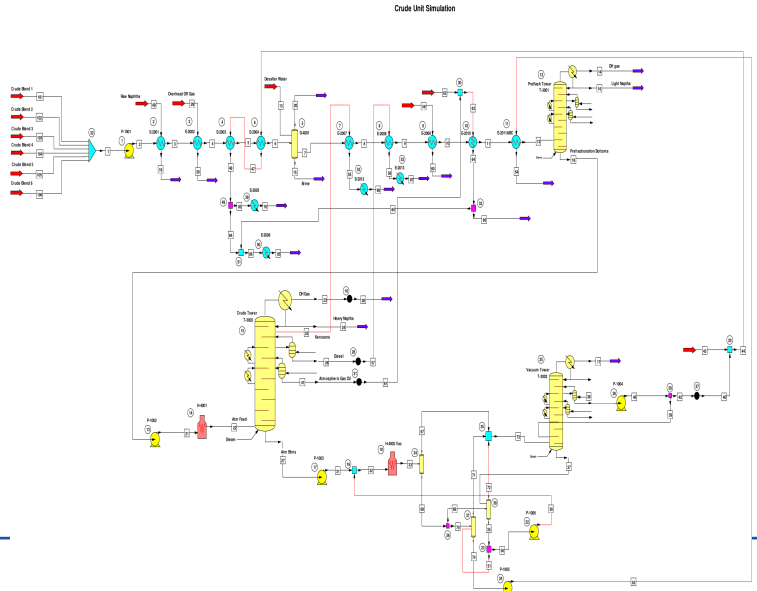




- Crowd simulation – to simulate large crowds, traffic estimations, social simulation, economic models are all based on agent based models
- Examine the behavior that lead to emergence
- Examine individual actions

Process Simulation

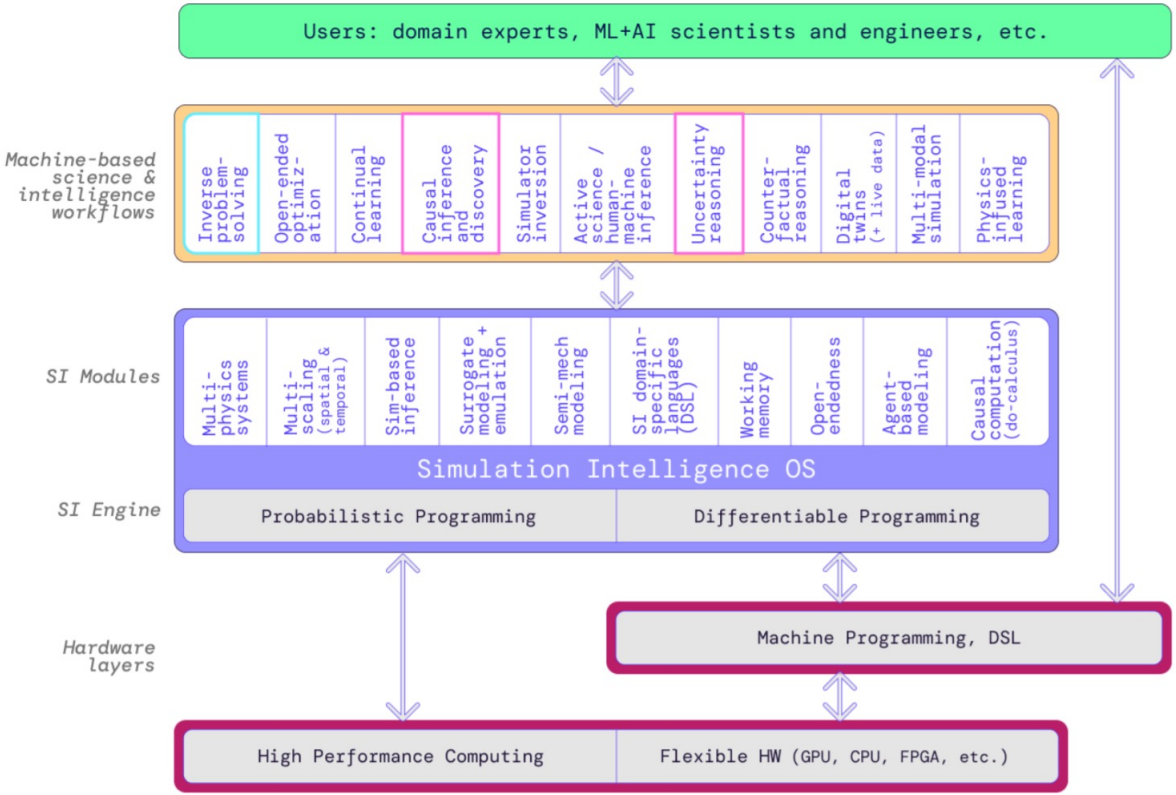
- Simulation of chemical, biological, industrial, ecological or medical processes



Hybrid

- Combinations of the above paradigms
- Most often, one paradigm is no longer sufficient
- Increasingly, merger of Simulations, Machine Learning and Scientific (High-Performance) Computing

For example ….


Users: domain experts, ML+AI scientists and engineers, etc.
Machine-based science & intelligence workflows
Inverse problem-solving
Open-ended optimiz-ation
Continual learning
Causal inference and discovery
Simulator inversion
Active science / human-machine inference
Uncertainty reasoning
Counter-factual reasoning
Digital twins (+ live data)
Multi-modal simulation
Physics-infused learning
SI Modules
Multi-physics systems
Multi-scaling (spatial & temporal)
Sim-based inference
Surrogate modeling + emulation
Semi-mech modeling
SI domain-specific languages (DSL)
Working memory
Open-endedness
Agent-based modeling
Causal computation (do-calculus)
Simulation Intelligence OS
SI Engine
Probabilistic Programming
Differentiable Programming
Machine Programming, DSL
Hardware layers
High Performance Computing
Flexible HW (GPU, CPU, FPGA, etc.)

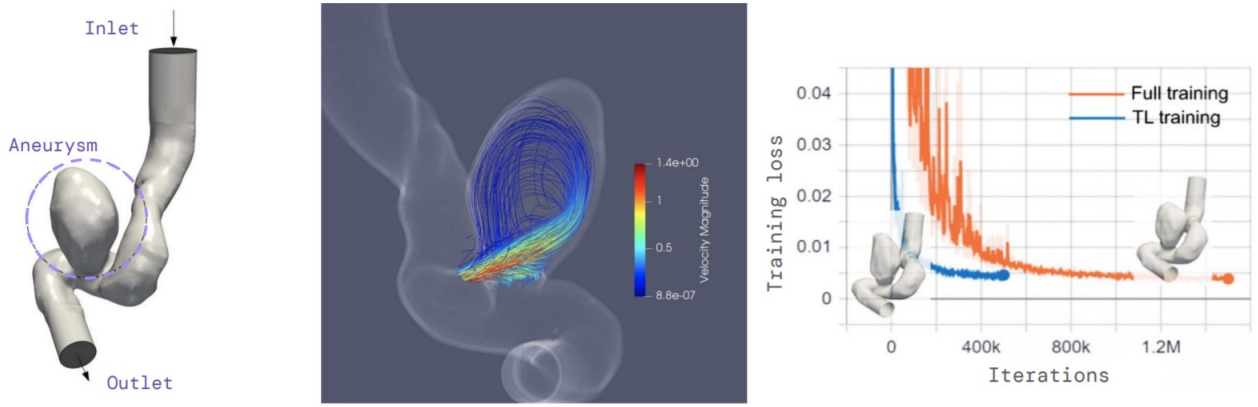


Figure 4: SimNet simulation results for the aneurysm problem [19]. Left: patient-specific geometry of an intracranial aneurysm. Center: Streamlines showing accurate flow field simulation inside the aneurysm sac. Right: Transfer learning within the NN-based simulator accelerates the computation of patient-specific geometries.

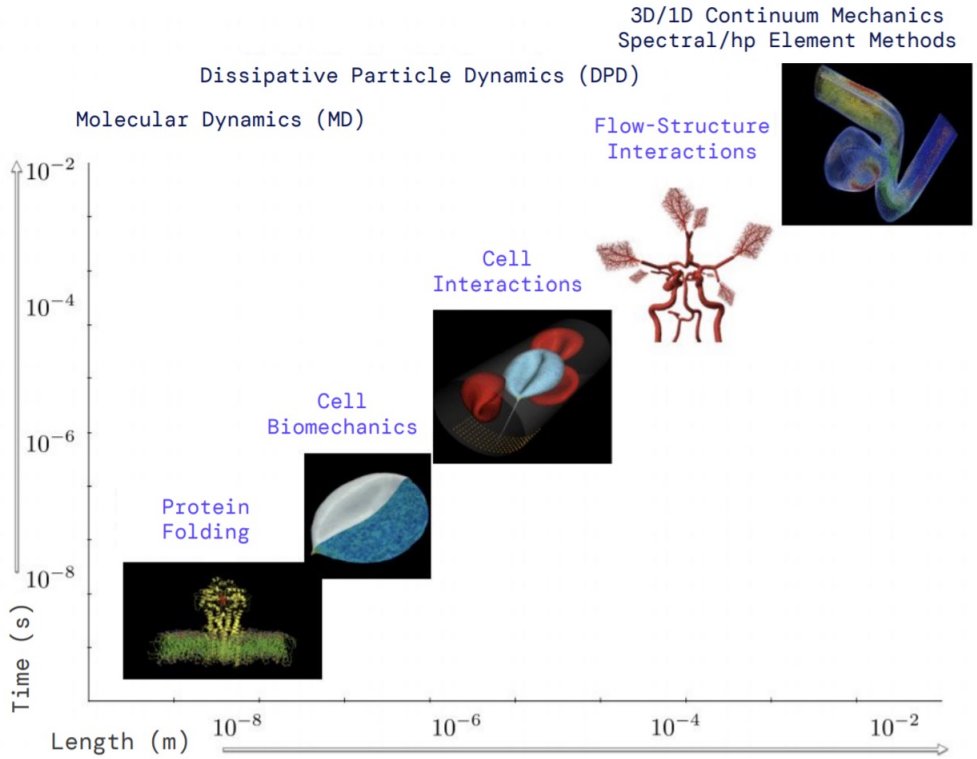


Figure 5: A cascade of spatial and temporal scales governing key biophysical mechanisms in brain blood flow requires effective multi-scale modeling and simulation techniques (inspired by [49]). Listed are the standard modeling approaches for each spatial and temporal regime, where an increase in computational demands is inevitable as one pursues an integrated resolution of interactions in finer and finer scales. SimNet [19], JAX MD [50], and related works with the motifs will help mitigate these computational demands and more seamlessly model dynamics across scales.

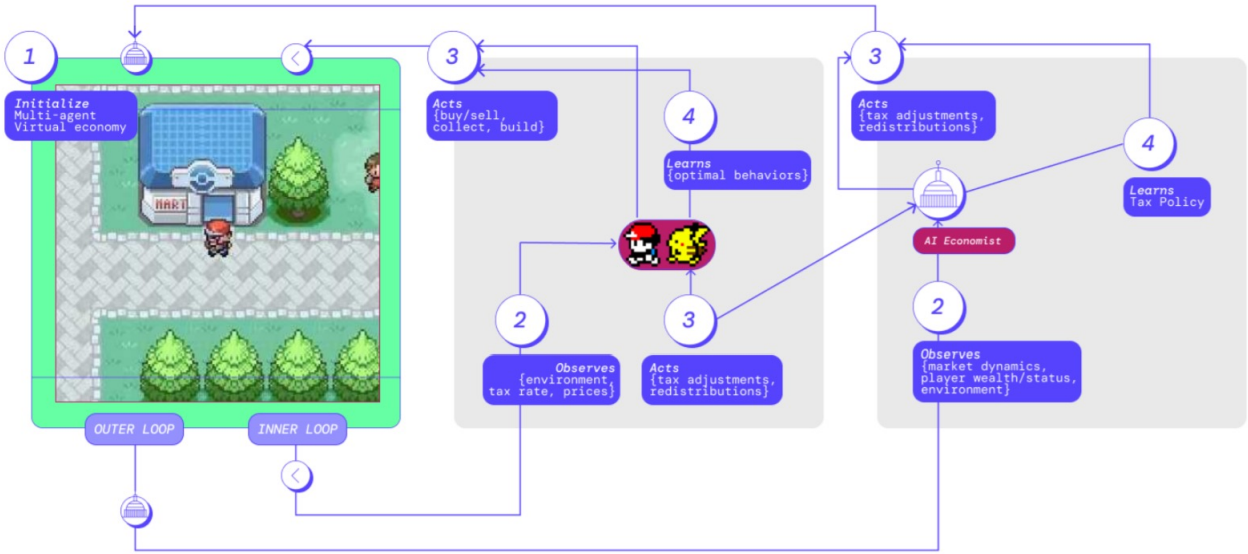
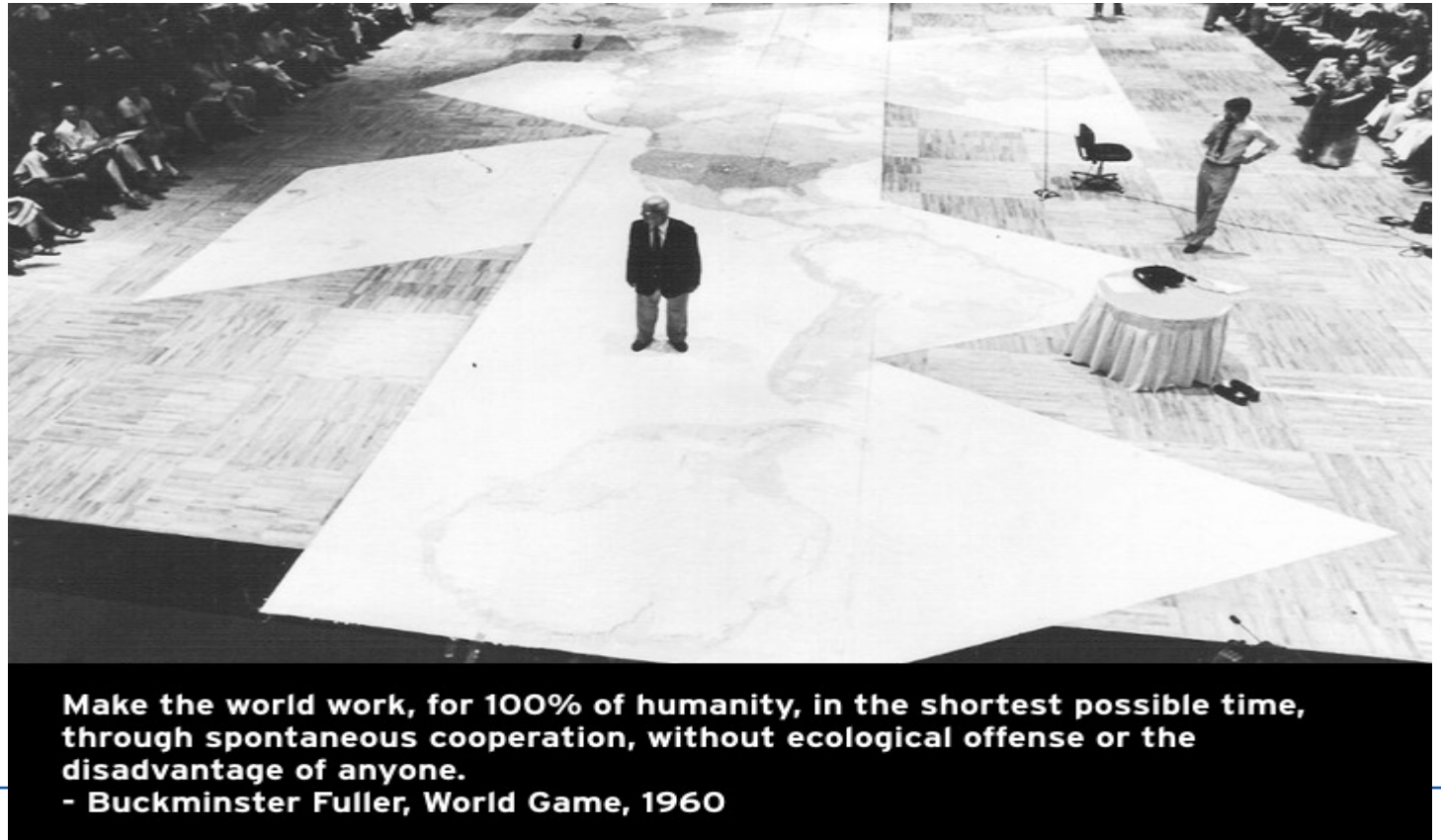


Figure 17: The AI Economist's two-level RL framework [10]. In the inner loop, various types of RL agents gain experience by performing labor, receiving income, and paying taxes, and learn through balancing exploration and exploitation how to adapt their behavior to maximize their utility. In the outer loop, the social planner adapts tax policies to optimize its social objective. The initial framework ran in a simplistic 2D environment with limited agency, whereas we show an example of work-in-progress extending AI Economist methods to more elaborate, complex multi-agent worlds [268].

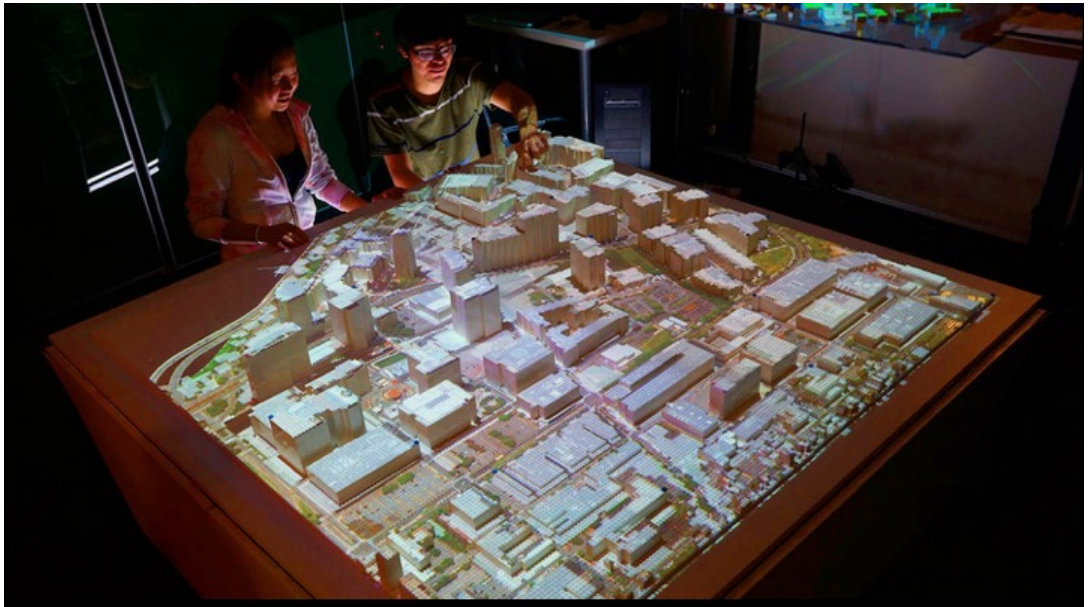
Human in the loop



Medical

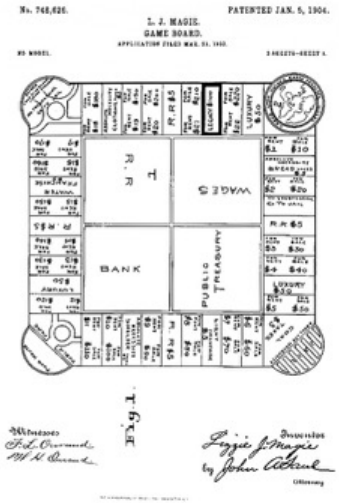


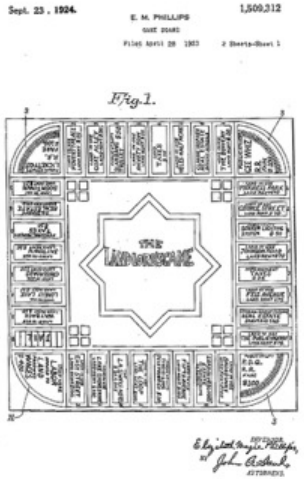
Simulation Gaming



"We develop simulation systems to predict and quantify the potential impact of disruptive interventions within new and existing cities. We place a special emphasis on augmented reality decision support systems (ARDSS) that facilitate non-expert stakeholder collaboration within complex urban environments. ""
MIT Intervention Simulation

Actual Games









What is a simulation? Redux

A simulation is a representation of a system, with some dynamics and process

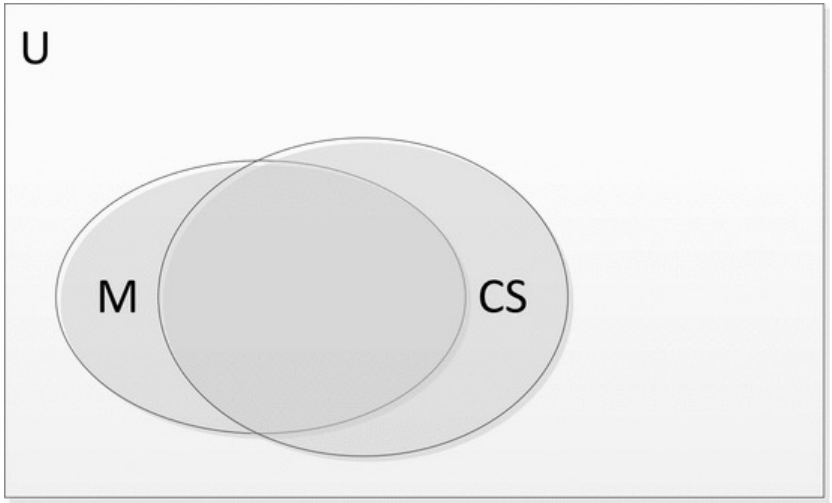
What does representation include and mean?

- Fidelity
- Validity
- Abstraction
- Resolution

Five Levels

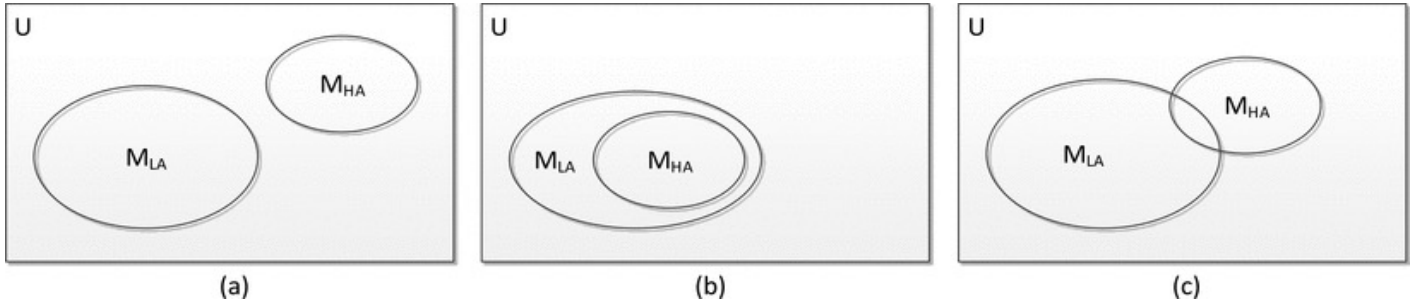
- the real system;
- the experimental frame;
- the base model;
- the lumped model;
- and the computer model.

Abstraction



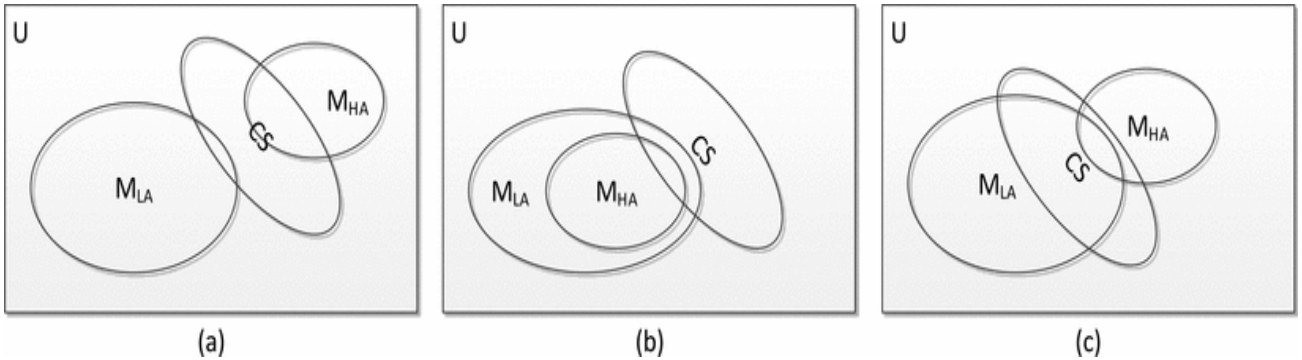
- U: Universe of Information on Reference System
- M: Model
- CS: Current Scenario

Resolution and/Or Detail



(a): The two models share no common information, and are abstracting different parts of the reference system.
(b): The two models focus on a common part of the reference system, and MLA contains more information in the modeled part of the reference system.
(c): Though there is some intersection, the two models focus on different parts of the reference system, and

Fidelity



(a): MLA covers less of the CS than MHA, therefore has lower fidelity
(b): MLA covers more
(c): Similar

Validity

- Event
- External
- Content
- Internal
- Educational
- Convergent
- Psychological

- Operational
- Face: Default, Efficient.

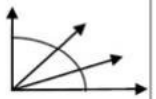
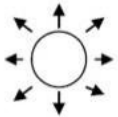


Abstraction of which object?

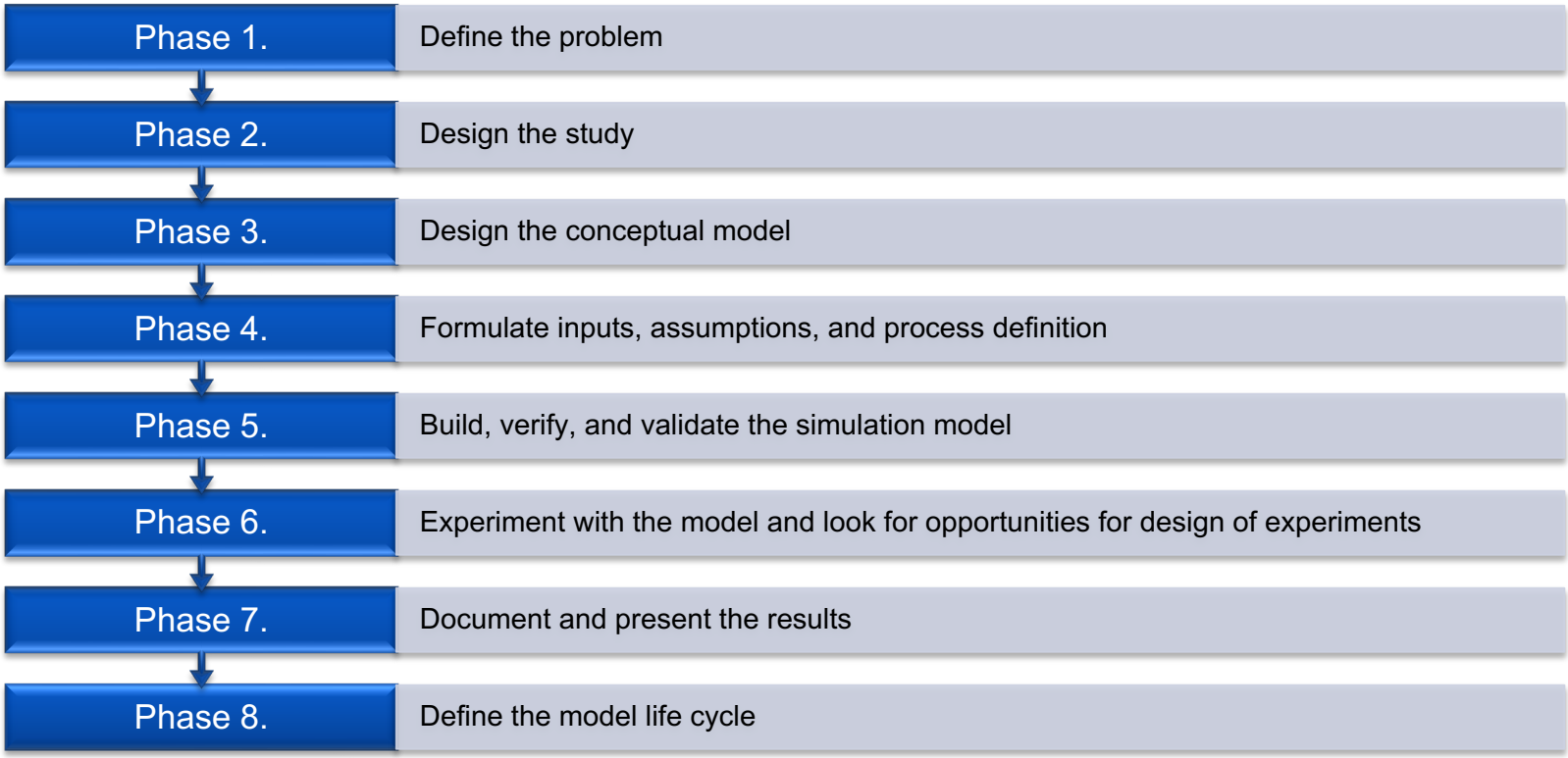
Resolution to which level?

Fidelity to how much realism?

Uncertainty

Complete Certainty		Level 1	Level 2	Level 3	Level 4	Total ignorance
	External forces (X)	A clear enough future (with sensitivity) 	Alternate futures (with probabilities) 	A multiplicity of plausible futures (unranked) 	Unknown future 	
	System Domain (R)	A single (deterministic) system model	A single (stochastic) system model	Several plausible system models, with different structures	Unknown system model; know we don't know	
	System Outcomes (O)	A point estimate for each outcome (with sensitivity)	A confidence interval for each outcome	A known range of outcomes	Unknown outcomes; know we don't know	
	Weights on outcomes (W)	A single set of weights	Several sets of weights, with a probability attached to each set	A known range of weights	Unknown weights; know we don't know	

How to build a simulation?



Phase 3: Conceptual Model

Step 1.	Decide on paradigm: continuous, discrete, or combined modeling.
Step 2.	Determine the elements that drive the system.
Step 3.	Determine the entities that should represent the system elements.
Step 4.	Determine the level of detail needed to describe the system components.
Step 5.	Determine the graphics requirements of the model.
Step 6.	Identify the areas that utilize special control logic.
Step 7.	Determine how to collect statistics in the model and communicate results to the customer

Typical Elements



Entities: represent active objects: ships, products, costumers, orders....



Attributes: information to describe entity *(different values for different entities)*



Variables: information to describe entire system *(same values for different entities)*

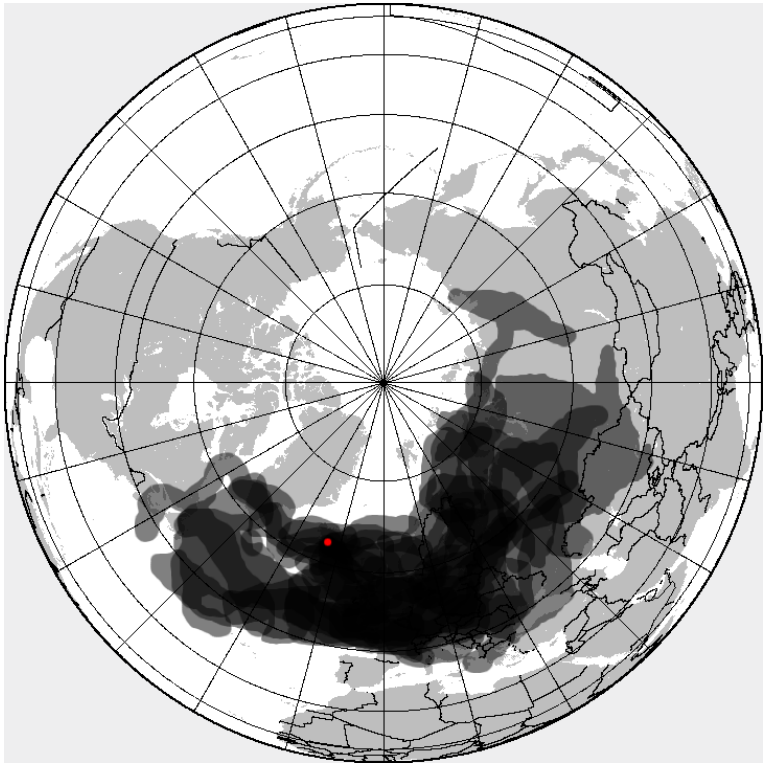


Resources: servers, machines, workers, nodes, links, runways, gates, agents, clerks....



Queues: place where entities wait for the resources

Exercise



Conceptual Models

- In 2010, Eyjafjallajökull volcano in Iceland erupted, causing a complete shutdown of all Air Travel across Europe and other parts of the Northern hemisphere from 15 till 23 April, and intermittent shutdowns later.
- Your task is to now design a simulation study that will:
  - Provide advice on safety for airplanes: cargo and passenger, jet and propeller
  - Provide a plan for rerouting of all stuck passengers, considering alternate modes of transport
- Define the problems, which problems can be addressed by which simulation methods
- Define inputs, outputs, uncertainties, human interventions (for example with rules, parameter values etc.)
- Define how and when the simulation(s) will be updated as the situation evolves